

Butter: A Density-Aware Adaptive Traffic Light System Using ML Based Vehicle Detection and Fuzzy Priority Scheduling

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Abstract

Urban traffic congestion is worsened by fixed-time traffic signals that fail to account for real-time variation in vehicle density across intersections. This project, Butter, proposes a traffic monitoring and adaptive signal control system that uses a deep learning-based object detector (YOLOv8) to estimate vehicle density at each approach of a four-way intersection, and dynamically allocates green-light duration using a custom-built fuzzy inference engine combined with weighted priority scheduling. The system is motivated by direct observation of real intersections where one approach experiences severe congestion while its cross-traffic counterpart proceeds with a fixed, unresponsive signal cycle regardless of actual demand.

Dataset: The system uses recorded traffic footage for vehicle detection, and the Simulation of Urban Mobility (SUMO) platform for controlled signal-timing evaluation. Given the limitations of general-purpose detection datasets (e.g. COCO) for Indian road conditions — specifically the absence of autorickshaws and other India-specific vehicle classes — future work includes fine-tuning YOLOv8 on the Bengaluru Mobility Dataset (BMD-45), a CCTV-perspective dataset annotated for Indian traffic.

Application: The core application is a real-time signal control simulator for a standard four-way intersection (North–South and East–West approach pairs, operating as two alternating signal phases). Given live or recorded footage from each approach, the system estimates per-approach congestion, aggregates congestion across each phase's paired approaches, and computes signal timing dynamically — aiming to reduce average wait time compared to a fixed-cycle baseline, with particular attention to preventing a single severely congested approach from being forced to wait out an unnecessarily long fixed cycle.

Methodology: (1) Vehicle detection using YOLOv8, a real-time single-pass object detector, per approach camera feed. (2) Lane/approach assignment via manually defined region-of-interest polygons and a point-in-polygon centroid test. (3) Density estimation, normalizing per-approach vehicle count and occupied bounding-box area into a single congestion score. (4) A custom-built Mamdani-style fuzzy inference engine converting congestion and elapsed wait time into a priority weight per approach. (5) Phase-level aggregation, taking the maximum of each phase's two constituent approaches' priority weights so a single congested approach can trigger an extended green phase. (6) Weighted priority scheduling with a minimum guaranteed time per phase. (7) Evaluation of average wait time and throughput against a fixed-time baseline using SUMO-simulated scenarios, with detection accuracy and control-algorithm effectiveness evaluated as separate, decoupled experiments.

Literature Survey

Adaptive traffic signal control has been an active research area for several decades, with approaches broadly falling into three categories relevant to this project: fuzzy-logic-based controllers, vision/deep-learning-based density estimation, and simulation-based evaluation frameworks such as SUMO.

Fuzzy logic control of traffic signals is well established, tracing back to foundational work in the late 1970s and maturing through the 1990s and 2000s as a technique for handling the inherent uncertainty in traffic measurements [10]. Early fuzzy controllers used simple inputs such as queue length and vehicle gap to adjust green-light duration [6], and later work extended this to four-way intersections explicitly designed for mixed traffic conditions, including a high proportion of two-wheelers [6] — a consideration directly relevant to Indian road conditions. Comparative work has since benchmarked fuzzy controllers against reinforcement-learning approaches such as Q-learning, generally finding fuzzy logic more interpretable and easier to tune with limited data, while RL approaches scale better across multiple coordinated intersections [5]. Neuro-fuzzy hybrids, which learn membership functions and rules from data rather than hand-tuning them, have also been explored as a way to reduce manual design effort, though at some cost to interpretability [9]. Other work has combined fuzzy logic directly with machine-learning-based detection in a single pipeline validated in SUMO [1], structurally similar to the approach taken in this project.

More recent work has shifted the detection layer from classical computer vision (background subtraction, blob analysis) toward deep-learning object detectors, primarily variants of YOLO, due to their robustness to occlusion, lighting variation, and camera angle compared to classical methods. Several studies use YOLO-based detection directly to drive adaptive signal timing based on queue density and per-vehicle waiting time [3], with more recent work (YOLOv11) demonstrating further improvements in detection accuracy and responsiveness for smart-city deployments [4]. Complementary work has focused specifically on improving density estimation itself, for example combining YOLOv8 with kernel density estimation to produce a more spatially precise congestion map than simple bounding-box counting [7]. System-level implementations combining a detection module, a signal-switching algorithm, and a lane-priority mechanism into one architecture [8] closely mirror the modular structure adopted in this project (detection → density scoring → priority inference → scheduling).

On the simulation side, SUMO has been used not only as a testbed for control algorithms but also in direct combination with vision-based tracking systems: one study built a YOLO-based vehicle and pedestrian tracking pipeline and incorporated its output into SUMO to reconstruct realistic turning-movement counts for intersection analysis [2], confirming that a vision-to-simulation bridge of the kind required by this project (given that SUMO's own simulated vehicles cannot be fed through a real-world-trained detector) is a viable and previously validated pattern.

Collectively, this body of work supports the overall approach taken in this project — combining a modern deep-learning detector with an interpretable fuzzy decision layer, evaluated in SUMO — while also indicating that reinforcement-learning-based coordination becomes the more common approach once multiple intersections must be coordinated jointly [5], which is noted here as a direction for future work beyond the current single-intersection scope.

Architecture Diagram

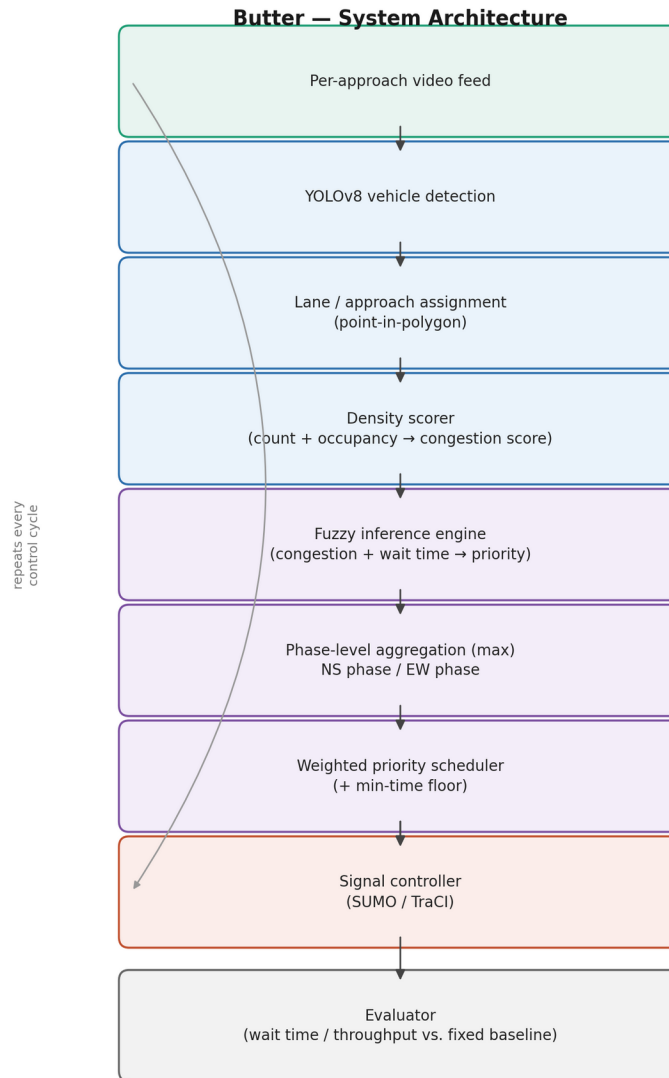


Figure 1: End-to-end system pipeline. Detection (blue) and fuzzy decision-making (purple) stages operate independently per approach each control cycle; phase aggregation and scheduling combine paired approaches (North–South and East–West) before signal timing is pushed to the SUMO/TraCI-controlled simulation and logged by the evaluator.

References

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